

Sequencing microsphere selective internal radiotherapy after external beam radiotherapy for hepatocellular carcinoma: proof of concept of a synergistic combination

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Abstract

Objectives: This study aims to explore the synergistic effects of combining stereotactic body radiation therapy (SBRT) and selective internal radiation therapy (SIRT) in that specific sequence for treating hepatocellular carcinoma (HCC), particularly in patients at high risk of radiation-induced liver disease (RILD).

Methods: We analysed a case of a patient with HCC who was treated with SBRT at our institution. A virtual ^{90}Y dose distribution was added using our in-house *MIDOS* model to keep a minimum dose to the healthy liver tissue. BED and EUD metrics were calculated to harmonize the dose distributions of SBRT and SIRT.

Results: Our radiation biology-based models suggest that the combination of SBRT and SIRT could maintain effective tumour control while reducing the dose to normal liver tissue. Specifically, an SBRT plan of 10 Gy $\times$ 3 fractions combined with SIRT yielded comparable tumour control probability to an SBRT-only plan of 10 Gy $\times$ 5 fractions.

Conclusions: The combination of SBRT and SIRT is a promising treatment strategy for HCC patients at high risk of RILD. While the LQ model and associated formalisms provide a useful starting point, further studies are needed to account for factors beyond these models. Nonetheless, the potential for significant dose reduction to normal tissue suggests that this combination therapy could offer substantial clinical benefits.

Advances in knowledge: This article presents a proposal to combine SBRT and SIRT, in this specific order, for HCC, discussing its advantages. A framework for future research into optimizing combination therapy for HCC is provided, utilizing a novel HCC vascular model to simulate ^{90}Y doses.

Keywords: SBRT; SIRT; HCC; liver cancer; radiation-induced liver disease; radiotherapy combinations.

Introduction

Liver cancer is the third leading cause of cancer mortality in the world, and its incidence is the seventh among the different types of cancer.¹ Worldwide, hepatocellular carcinoma (HCC) is the most common type of liver cancer, accounting for around 75% of all liver cancers.² To treat HCC, liver surgery remains the gold standard curative therapy, although this option is restricted by underlying liver malfunction and concurrent medical complications.³ Radiation therapies, such as external beam radiation therapy (EBRT) and selective internal radiation therapy (SIRT), may play an important role in HCC treatment.^{3,4} In patients with unresectable and small HCC, staged as intermediate (stage B under the Barcelona Clinic Liver Cancer [BCLC] staging system) or even early disease (stage A) when resection, transplantation, and local ablation are not an option. Transarterial chemoembolization (TACE) is the standard of care for these patients, with the role of EBRT usually limited to alternative therapy or as a bridging/downstaging strategy—and also selective internal radiotherapy (SIRT)—due to the concern of increased toxicity when irradiating large volumes.

The development of intensity-modulated radiotherapy in EBRT transformed this modality into a treatment choice for unresectable HCC. In fact, better clinical outcomes with this technique have been observed through stereotactic body

radiation therapy (SBRT) and hypo-fractionation, allowing accurate delivery of high uniform doses to the tumour in a few fractions while sparing the normal tissue.^{3–5} However, liver toxicity remains a dose-limiting factor affecting ~25% of the patients.⁶ This is due to differences in radiosensitivity across patients with normal and cirrhotic livers, variability in the pathological characteristics of liver tumours, and SBRT-related toxicity, such as gastrointestinal injury and central hepatobiliary tract toxicity.^{4,7} In contrast, SIRT has the advantage of selectively administering the dose to the tumour, avoiding normal liver toxicity due to the dual blood supply between tumours and normal parenchyma.^{8,9} Nevertheless, SIRT leads to a heterogeneous yttrium-90 (^{90}Y) dose distribution, resulting in cold spots in the tumour.^{10,11} Given the complementary aspects of both approaches, combining both therapies is growing in interest to improve patient outcomes.^{12,13}

The main concern about combining SIRT and SBRT is predicting outcomes, which is not as straightforward as when analysed individually.¹⁴ While it is well known that the patient outcome to SBRT and SIRT is related to the absorbed dose,^{15–18} that is not obvious when combining them up since the correlation between dose and effect varies modality-wise.¹² Differences between modalities reside in the dose-pattern uniformity and the delivery time structure. For liver

SBRT, the dose is uniformly delivered at a high-dose rate in the target with steep dose gradients to spare normal tissue.^{4,5} However, SIRT leads to a nonuniform ⁹⁰Y dose distribution in the target, mainly depending on the tumour vasculature, with an exponential time structure related to the ⁹⁰Y radioactive decay.^{9,19} Therefore, the doses administered under these modalities are not radiobiologically equivalent.^{12,14} Radiation effect models provide estimations to explore potential combinations. The linear quadratic (LQ) model has been used to evaluate a combination of SIRT and SBRT addressing the biologically effective dose (BED), the equivalent dose (EQD), and the equivalent uniform biological effective dose (EUBED).²⁰⁻²² Retrospective patient data from SBRT and SIRT were used to evaluate BED, EQD, and EUBED metrics. Indeed, a prospective phase 1 trial is currently being conducted, where patients are treated with SBRT following SIRT when lesions are potentially underdosed and then nonresponding after SIRT.²¹

In this work, we present a proof-of-concept to determine dose sizes in combinations of SIRT following SBRT. As opposed to other proposed combinations,²⁰⁻²² we propose to use SBRT to give the tumour as much dose as possible, considering the limitation in the maximum dose to the normal liver. Then, SIRT will play the role of a “boost” by compensating for the lack of dose delivered from SBRT for those patients with a high risk for liver toxicity. We, therefore, aim at the maximum possible activity administered in SIRT while keeping the normal parenchyma dose negligible. The strengths and limitations of the LQ model to estimate the effects of this combination therapy are discussed, as well as the potential patients who might benefit from this approach in the design of a clinical trial.

Methods

Outline of the proposed framework for the SBRT-SIRT combination

One patient with an HCC lesion treated with SBRT (5 fractions $\times$ 10 Gy) at the Massachusetts General Hospital was retrospectively selected as a model to develop the optimal way to combine SBRT and SIRT. In this manner, we took the original SBRT plan as a base for our analysis. Then, we added a virtual ⁹⁰Y dose distribution with negligible dose to the normal parenchyma. To this end, SIRT was simulated using our *MIDOS*²³ model to estimate the optimal number of microspheres perfused to lead to a marginal dose in the normal liver. Then, these microspheres were spatially distributed in the tumour according to an in-house computational tumour vasculature model, and TOPAS²⁴ was used to perform the SIRT dose calculation. Finally, BED and EUD metrics were calculated to harmonize SBRT and SIRT dose distributions.

MIDOS: a model to optimize microsphere administration in SIRT

To estimate the optimal number of microspheres administered (ie, activity perfused) to the tumour while keeping a negligible dose to the liver, a SIRT treatment was simulated by an in-house developed model, *MIDOS*.²³ Our model uses adult male and female liver vasculature phantoms²⁵ to predict the amount of activity delivered to the tumour and normal hepatic parenchyma, given the desired activity to perfuse, the type of microsphere, and the catheter's injection point, among other factors. Perfusion is modelled through the hepatic artery following the

conservation of matter, Poiseuille's and Murray's laws. Finally, *MIDOS* includes a tumour uptake model depending on the grade of tumour vascularization. In this way, the tumour fills out with microspheres until it reaches saturation, and the next microspheres will flow to the normal liver, where they get lodged. In this context, *MIDOS* was used to predict the optimal number of ⁹⁰Y (TheraSphere) to treat a spherical tumour placed at the sixth segment of the adult liver male phantom. The tumour volume was obtained from the SBRT plan carried out on our patient model. More details on the *MIDOS* model can be found elsewhere.²³

Of note, it has been observed *ex vivo*^{10,26,27} that microspheres tend to form clusters in the blood vessels. This effect was included in the *MIDOS* model using the data reported by Pasciak et al.²⁷ Therefore, the output of *MIDOS* was the optimal number of microspheres in the tumour, that is in turn the input of our in-house computational tumour vasculature to distribute them spatially.

A computational vasculature and virtual dose distribution in HCC

A computational tumour vasculature model was developed to distribute the clusters of microspheres in the tumour obtained from the *MIDOS* model. The model is a simplified version of AMBER (Agent-Based Model for Biophysical Evolution after Radiotherapy).²⁸ AMBER includes cells, nutrients, and extracellular factors as agents and interaction rules among them that contribute to generating tumour vasculatures, as shown in Figure 1. To grow the vasculature, we initially set an arbitrary VEGF gradient map and pressure map within the gross tumour volume contoured on the patient. Each vessel started from a point outside the tumour, grew towards it, and split into smaller vessels stochastically. The growth of the vessel resulted from a directed random walk towards a high VEGF concentration and the effect of the pressure, using mechanical pressure as a barrier by having an antagonist effect on the VEGF gradient until reaching a high-pressure threshold to deter vessel growth.

We created 500 vessels starting at random positions on a sphere outside the tumour surface and let them grow following the previously described algorithm. Once the full vasculature was generated, we used the number of microspheres determined by *MIDOS* to create clusters, according to a specific tumour-cluster pattern, represented as a log-normal distribution.²⁶ We, therefore, created clusters by sampling this

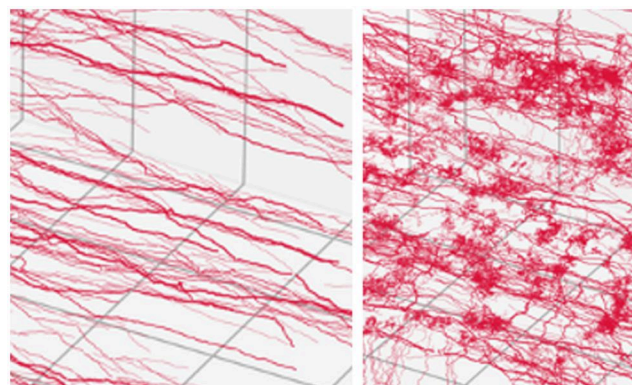


Figure 1. Examples of the vasculature generated by our AMBER model for normal tissue (left) and tumour (right).

distribution until reaching the total number of microspheres in the tumour determined by *MIDOS*.

Microspheres were injected into the tumour from a particular direction to mimic real radioembolization scenarios. Therefore, vessels close to the injection area received more microsphere clusters than vessels far away from the injection. Microsphere clusters were allowed to flow through the tumour vascular tree following the same principles as for *MIDOS*. Their probability of getting stuck at a given point was assumed to be proportional to the ratio of the microsphere (assumed spherical with a 12.5- μm radius) and the vessel area.

Using the information on cluster size and position, we used the TOPAS Monte Carlo toolkit²⁴ to simulate the radiation dose delivered to the tumour. We set up a configuration file that used the information on each Y90 cluster size and coordinates of the location where it got stuck and simulated its irradiation on the imported DICOM file of the patient. We specified an initial activity²⁹ of 2.5 MBq for each of the 24 066 microspheres that created 5360 clusters. By simulating the radioactive decay over a period of 22 half-lives, a fraction of only $2.38 \cdot 10^{-7}$ of the original activity remained and obtained the accumulated dose.

Combining doses through the BED and EUD formalism

The time structure of a given treatment plays a prominent role in the response to radiation. SBRT involves a small number of high-dose irradiations delivered in minutes, while SIRT delivers radiation continuously at a lower dose rate over the span of days or weeks. Therefore, the absorbed dose is not a quantity appropriate to predict or compare the effects of these treatment modalities. In the context of the LQ model, it is possible to harmonize physical quantities leading to the same effect through the concept of BED, which accounts for differences in the irradiation timing by modifying the β factor in the LQ model through the so-called Lea-Catchside factor.³⁰ BED can be calculated simply for both fractionated schemes and monoexponentially decaying dose rates, as is the case for SIRT. We obtained BED voxel-by-voxel in the tumour for the SIRT treatment using an α/β ratio of 13.4 Gy, corresponding to HCC tumours as shown by van Leeuwen et al.³¹

Additionally, typical dose distributions from SIRT are highly nonuniform, with hot spots near concentration of microsphere clusters and cold spots in poorly vascularized areas. This may lead to clonogenic cells in the tumour being underdosed, which can hamper the effectiveness of the treatment in controlling the tumour. Assuming all the voxels in the tumour contain the same number of clonogenic cells, it is possible to determine the equivalent uniform dose (EUD) that would lead to the same clonogenic response in the entire tumour as the considered SIRT nonuniform distribution.³² This same concept can be applied to the voxel-by-voxel calculated BED, obtaining the equivalent uniform BED (EUBED), defined by Niemierko as:

$$\text{EUBED} = \left(\sum_i v_i \cdot \text{BED}_i^{\frac{1}{\alpha}} \right)^{\alpha}$$

where v_i is the volume of the voxel i , BED_i is the i th-voxel BED, and α is the parameter for the linear component in the

LQ formalism, according to the original formulation of EUBED.³¹ Of note, the *generalized* EUD, was proposed to include effects in normal tissues, not driven necessarily by cell killing, by generalizing the value of α . However, using the original EUD formulation for tumour control, assuming it is mediated by cell killing, should still provide reasonable results from the radiobiological point of view. We therefore employed 0.037 Gy^{-1} for the α value according to van Leeuwen et al.³¹ as the working value for subsequent analysis. Also, we studied the sensitivity of EUBED to the value of this parameter.

For the SBRT treatment, we assumed the dose is designed to be uniform in the tumour, and therefore we calculated the BED corresponding to the prescribed dose (10 Gy $\times$ 5 fractions) using the same α/β ratio as above. As dose uniformity was assumed, EUBED was considered equal to this value for the SBRT modality.

Determining the optimal dose from EBRT

We calculated BED corresponding to the SBRT treatment alone, which was considered the reference scenario in our study. As explained in previous subsections, we also calculated the EUBED corresponding to our SIRT treatment. In order to match the effect on the tumour through the combination of both techniques, we studied how many fractions of SBRT with 10 Gy/fx and 5 Gy/fx would provide the necessary effect to match the reference scenario when added to the EUBED provided by the SIRT modality. This approach illustrates an operational framework to decide how to combine both modalities.

Results

Figure 2 shows the distribution of microspheres within our computational vasculature in the tumour. As the vasculature does not reach the centre of the tumour due to the effect of physical pressure, no microspheres can reach deep into the tumour. Microspheres form a shell around the tumour core in its periphery. Furthermore, the microspheres are not equally distributed, as more clusters tend to lodge closer to the point of injection, as shown in the top right panel of Figure 2.

Dosimetric results for this distribution of microspheres are illustrated in Figure 3. We calculated the absorbed dose with TOPAS, then obtained voxel-by-voxel BED values in the tumour, and finally obtained EUBED using the resulting distribution. We analysed the dependence of EUBED as a function of the α parameter in the Niemierko equation, showing a considerable increase in the predicted effect if the α value is estimated below the reference value selected in this work, $\alpha = 0.037 \text{ Gy}^{-1}$. With this value, the SIRT plan we produced contributed with an $\text{EUBED} = 39.1 \text{ Gy}_{\text{BED}}$ to the tumour effect.

We calculated the BED corresponding to an SBRT plan of 10 Gy $\times$ 5 fractions according to the LQ model, yielding a value of $87.9 \text{ Gy}_{\text{BED}}$. To match this value and keep the same effect in the tumour when adding the SIRT effect, 10 Gy $\times$ 3 fractions would be necessary, as this combination yields $52.7 \text{ Gy}_{\text{BED}}$. Figure 4 shows the dose distribution from the SBRT plan for 3 fractions of 10 Gy. As also shown in Figure 4, if 5 Gy per fraction would be preferable, 7 fractions would be necessary. This result generally illustrates an operational procedure to obtain the necessary remaining effect from the SBRT treatment once the optimal contribution from SIRT is

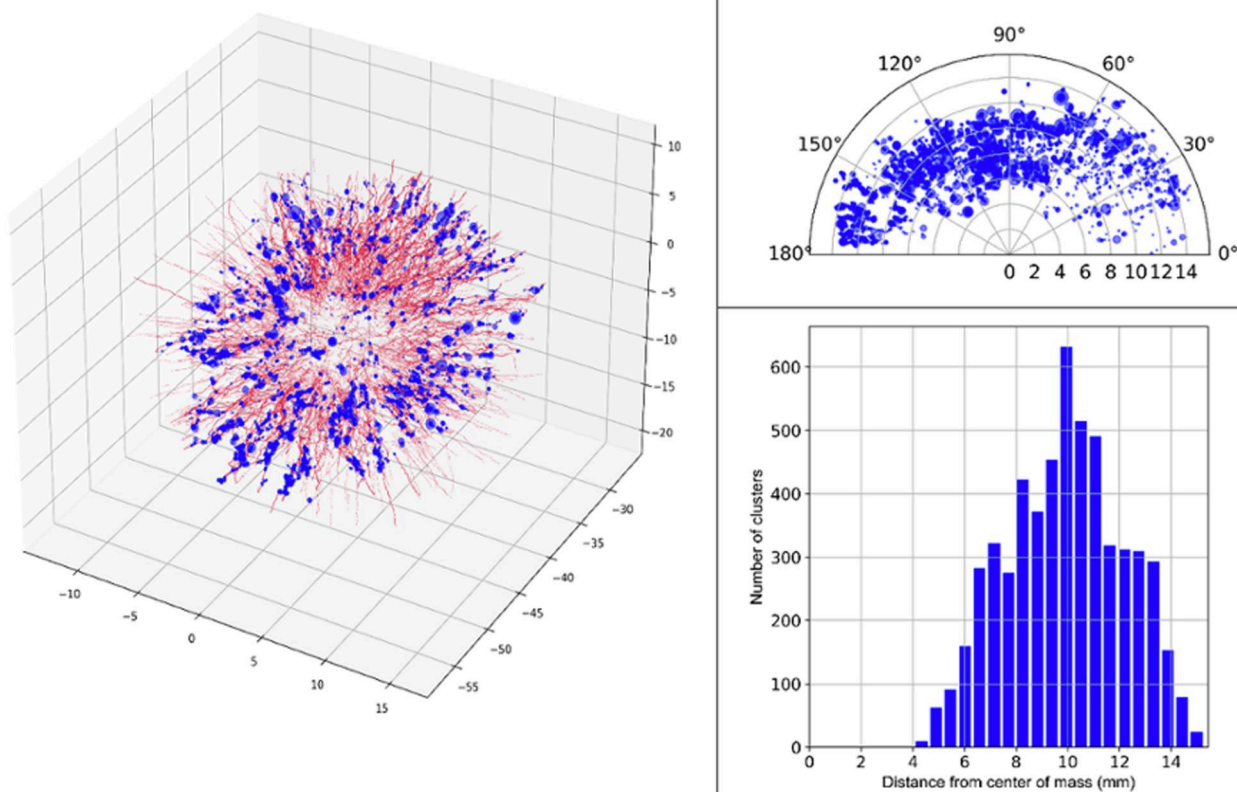


Figure 2. Microsphere distribution in the tumour. The left panel shows the generated tumour vasculature in red along with the microsphere clusters as blue dots. The dot size is proportional to the number of microspheres in the cluster. The top right panel shows the clusters in the tumour on a polar plot, the angle is computed in reference to the direction of injection and the radius is the distance of each cluster to the centre of the tumour. Injection comes from the left side of the plot. The bottom right panel shows the distribution of microsphere clusters as a function of distance to the tumour centre mass.

determined. Finally, Figure 4 also shows the contribution of both modalities to tumour control using the tumour control probability (TCP) curve for liver cancer determined by Lausch et al.³³ as a function of BED. Interestingly, the portion of the BED contributed by the SIRT treatment is, in principle, exempt when doing a similar analysis for hepatic non-TCP (NTCP).

Discussion

Radiation-induced liver disease (RILD) is a severe concern for most patients with unresectable HCC treated with any form of radiation-based therapy. Selective therapies that minimize the dose received by the normal parenchyma pose the opportunity to mitigate this effect but at the potential cost of falling short of the necessary dose to ensure tumour control. In this work, we show the rationale for a synergistic combination of SBRT, which provides a conformal dose to the entire target, and SIRT, which can help reduce doses to the liver parenchyma and, hence, RILD incidence. This is especially relevant for patients at high risk of RILD because of cirrhosis or, in general, poor liver function at treatment baseline. However, how to specifically design this combination, both in terms of dosage and sequencing, remains unexplored, as it is necessary to rely on radiation effect models before direct clinical observations are available.

HCC is a highly vascular tumour that receives the blood supply mainly through the hepatic artery, as opposed to the liver parenchyma that is supplied by the portal vein. This is the basis of SIRT, where microspheres are perfused into the tumour-feeding arteries to occlude them and irradiate the tumour internally. Due to this differential uptake, it is assumed that a tumour behaves as a sink of microspheres before they start lodging in the normal tissue vasculature, providing the opportunity of a considerable dose to the tumour at little cost in terms of parenchyma dose. This effect is backed by *ex vivo* studies,^{10,26} although the optimal point for tumour uptake will likely vary patient- and tumour-wise. Therefore, novel methods such as our MIDOS simulator,²³ or direct estimations from SPECT scans with ^{99m}Tc-MAA perfusions would be required for the combination design.

This work provides a methodology based on the LQ model and the BED and EUD formalisms to estimate effects in the tumour when combining external radiotherapy and SIRT. This methodology contains a series of assumptions. First, in the BED formalism, each voxel contains independent clonogenic cell populations, which survive to a given dose in a linear quadratic fashion and according to the same α/β ratio. Second, it is assumed that SBRT and SIRT are assumed to provoke equal effects on cells if they were delivered in the same time pattern, disregarding differences produced by radiation quality or effects on the tumour microenvironment,

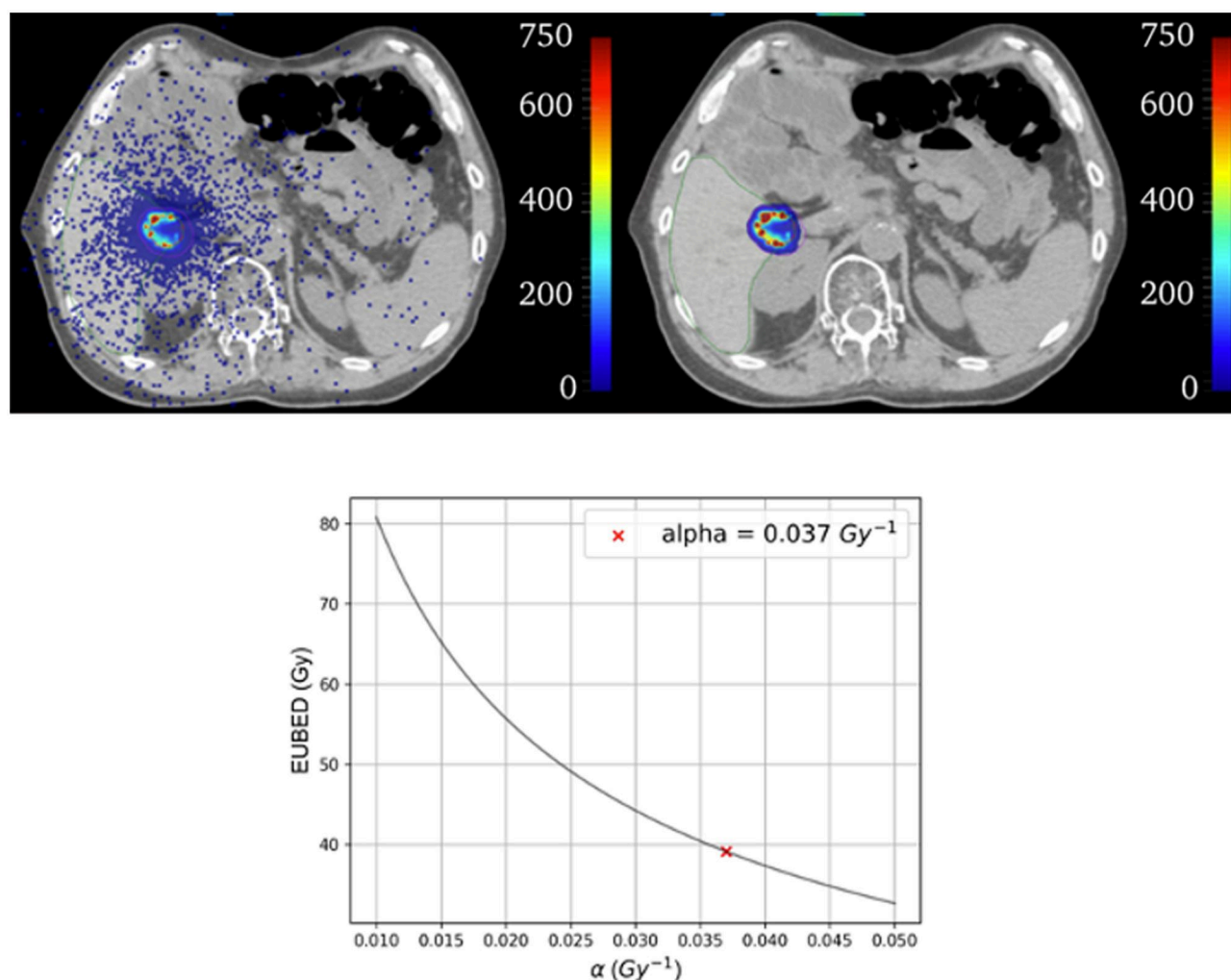


Figure 3. Dose distributions the Selective Internal Radiation Therapy plan (top left panel), with a bar showing Gy units (physical absorbed dose). The top right panel shows BED calculated for the showing the bar Gy_{BED} units. The bottom panel shows the sensitivity analysis of EUBED calculated using the nonuniform BED distribution in the tumour (contour in the top panels). The cross shows the picked value of 0.037 Gy^{-1} . Abbreviations: BED=biologically effective dose, EUBED=equivalent uniform biological effective dose.

particularly tumour vasculature. Third, the EUD formalism considers each voxel equally important for tumour control. Finally, sequence-derived effects are disregarded, considering that the tumour response to each modality is independent of the effects induced by the first therapy in place. While these assumptions are most likely unrealistic, models provide an easy-to-use tool to roughly estimate the necessary dosages from both therapies to achieve a similar tumour response as reference cases, such as SBRT alone. However, further consideration of these 4 simplifying assumptions is an essential element to study both preclinically and clinically, designing strategies to assess their importance in HCC cases. Additionally, even when these conditions are met, the values of α/β and α parameters, which are improbable to be accurately measured in the clinics and are likely patient-specific, strongly influence the calculated BED and EUBED, as shown in Figure 4, making calculations uncertain. In particular, the parameters employed in our study are known to be subject to inter (and intra-) patient variabilities so that values would change in each case. However, this is the case for applying SBRT and other radiobiology-driven operational procedures in radiation oncology, such as treatment compensation for missing fractions. This, in fact, supports the suitability for

presenting this case with a proof-of-concept to assess whether the order of magnitude in dose is reasonable or out of the clinical possibilities.

The present study has several limitations, as it is virtual without incorporating clinical data, our dosimetry is illustrative, and no clinical outcomes are included. However, the objective of this article is to show a proof-of-concept for this combination, showing the feasibility of achieving reasonable doses from each treatment modality and, particularly, that the heterogeneous doses that are characteristic of SIRT can be compensated with SBRT according to radiation biology principles. In this study, we focused on a patient meeting the clinical conditions of interest to maximize the benefit of this combination. It used a clinically delivered SBRT plan and reproduced a well-known pattern of heterogeneous dose to a well-vascularized tumour from SIRT, showing that reasonable tradeoffs between NTCP and TCP might be achieved with relatively few microspheres. This is a required condition to avoid spilling out of radioactivity into the healthy liver, ie, high tumour-normal ratio. To achieve this scenario, we created an optimized SIRT plan on a single and well-vascularized tumour lesion located in a specific segment. Another limitation is the tumour-cluster pattern

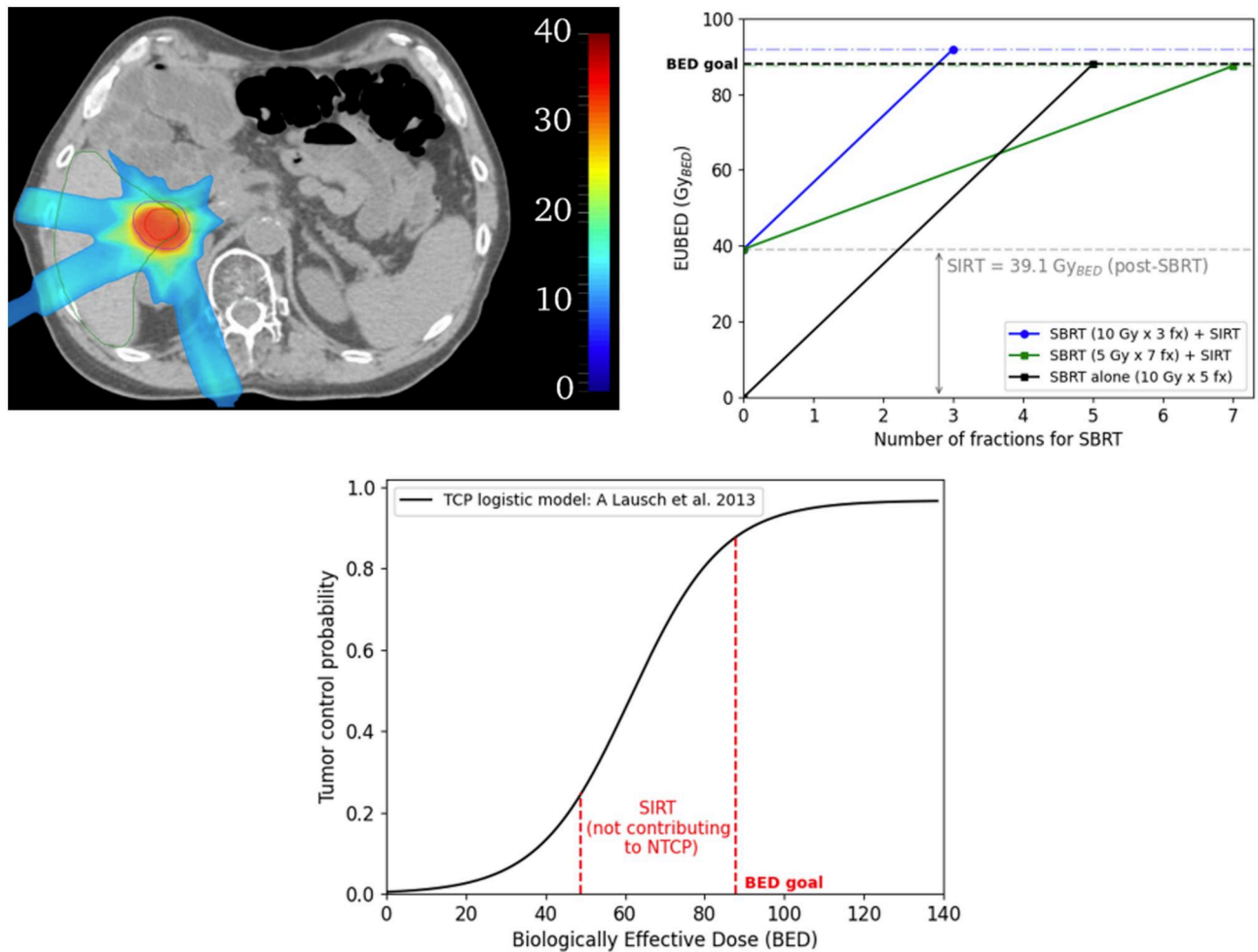


Figure 4. [Top left] Dose distribution for the SBRT plan with 3 fractions of 10 Gy. The colour bar shows Gy units. [Top right] Equivalent uniform biological effective dose as a function of the number of fractions to be delivered in the SBRT plan once the selective internal radiation therapy contribution is fixed, as determined by the optimal tumour-to-normal ratio with negligible dose to the healthy liver. Two fractionation schemes (10 Gy/fx and 5 Gy/fx) are compared. [Bottom] Tumour control probability as a function of biologically effective dose and contributions from each one of the proposed modalities. Abbreviation: SBRT=stereotactic body radiation therapy.

incorporated, which only analysed resin microspheres, whereas we present a case with glass microspheres. In this sense, further human histological studies including different types of microspheres should be carried out to improve the liability of the results. We also declined to utilize patients treated for SIRT in our institution since most have lobar or multifocal diseases with large extensions of ^{90}Y -microspheres. Additionally, scaling down the dose in SBRT or EBRT, in general, is immediate as it requires decreasing the fluence on the employed beams, whereas reducing the number of microspheres injected completely changes the spatial distribution of these, so that optimization is no longer available.

Combinations of SIRT and SBRT have been previously proposed. Wang et al.²⁰ studied the effects of sequential EBRT as a salvage treatment in 22 patients after incomplete response from initial SIRT (12/22), portal vein thrombosis (11/22), and/or lymphadenopathy (4/22). Patients treated initially with SIRT had either locoregional or locally advanced diseases with multifocal HCC that are refractory to chemoembolization or systemic therapies. Also, SIRT is performed through segmental, lobar, or even whole-liver treatment, which increases the radiation dose to the parenchyma and

deviates significantly from the idea proposed in this article. There is also a study on how to produce SBRT dosimetry plans to compensate for the initial SIRT map for this scenario, such as the one by Mee et al.²¹ Similarly, Abbott et al.¹² performed dosimetric analyses with the BED formalism and reported survival for 10 patients with advanced HCC treated with SIRT and palliative EBRT at the Sarah Cannon CC (Nashville, TN). However, these approaches still differ from ours in 2 key aspects: indications and combination sequence.

We identify synergistic components in SBRT and SIRT and think they can address the 2 main clinical concerns for intermediate HCC: tumour control and liver preservation. We also propose that it has the potential to be optimal in a specific order and not the other way around. Prior radioembolization of tumour vasculature might lead to hypoxia in the tumour microenvironment, ie, insufficient oxygenation in the tumour.^{34,35} Although studies have shown that SIRT is effective in treating HCC, tumour recurrence, and poor prognosis have been related to the effect of hypoxia.^{36,37} Hypoxic tumours have a poor response to radiotherapy as they are more radioresistant.³⁸ Thus, the effectiveness of combining SBRT following SIRT might not reach its full potential, deviating from known tumour response-

dose relations observed in SBRT alone. Conversely, if SBRT is initially imparted, it might impact the vasculature tree of both liver parenchyma and tumour,^{39,40} effectively modifying the potential dose distribution of SIRT. High perfusion values in embolization procedures induce tumour hypoxia, mitigating the effectiveness of subsequent radiation therapy to some extent. The opposite is not true, and according to the review by Park et al.,³⁹ moderately high doses of instantaneous radiation (5-10 Gy) make tumour blood flow increase for 2-3 days, which could be utilized to improve perfusion of microspheres to the tumour.⁴¹⁻⁴³ Additionally, time gaps between therapies can lead to tumour repopulation and vascular regrowth, enlarging the uncertainties associated with the combined effects and providing an opportunity to regularize the tumour vasculature before radioembolization. It must be noted that cell-killing-based radiobiological models do not capture these effects, and they rather provide a rationale to explore this specific sequencing clinically. McGee et al.⁴³ explored this specific order, but only for patients with advanced HCC and gross vascular invasion. In this scenario, SIRT delivered a higher dose to intravascular tumours than primary intermediate-stage HCC. Their results showed an encouraging increase in the survival probability for patients receiving SIRT after SBRT compared to SBRT alone. This might partially be due to the complementary effect in the primary tumour, supporting our idea of SBRT followed by SIRT as a curative option for earlier stages of HCC.

The ideas summarized in this work are intended to apply primarily to the design of clinical trials. Patients at high risk of RILD and highly vascularized tumours might benefit from this combination therapy. Using SBRT first, a bedrock uniform dose can be provided in the tumour without exceeding tolerance limits in the normal tissue. Then, SIRT can be performed due to the high grade of tumour vascularization, which is translated into a high uptake of ⁹⁰Y microspheres in the tumour, sparing the liver parenchyma. However, the optimal activity perfused in SIRT needs to be previously estimated, for which we used our in-house MIDOS model. In such scenarios, combining both therapies appears synergistic, reducing the dose delivered to the healthy tissue while maintaining TCP.

Conclusions

This article proposes a new therapeutic option for these patients, designed and conceived from radiobiological principles as a synergistic therapy from the beginning and not because of a lack of response or salvage from further complications. The synergistic combination of SBRT and SIRT presents a promising avenue for treating patients who are at high risk of RILD with highly vascularized tumours. Our study leverages the LQ model, along with BED, EUD, and TCP formalisms, to provide a preliminary framework for optimizing this combination therapy. While these models offer valuable insights into dose optimization and tumour control, they are limited. Factors such as tumour hypoxia, vascular changes, and radiobiological parameters necessitate further preclinical and clinical investigations. Nevertheless, our initial findings suggest that this combination therapy could substantially reduce the dose to normal liver tissue while maintaining effective tumour control, thereby offering a potential clinical benefit for high-risk patients.

Funding

This work was supported by the NIH/NCI (award no. R00CA267560).

Conflicts of interest

The authors have no conflicts of interest to declare.

Data availability

The data generated for this study are based on computational models and simulations. No clinical data was used. The computational models, including the parameters and algorithms, are available from the corresponding author upon request.

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